Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 7	**Date: 015-08-2026**	**Enrolment No:92400133079**

Code 1 :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image directly in grayscale
src_image = cv2.imread(r"D:\SEM 3\python\IMG_2239.JPG",
cv2.IMREAD_GRAYSCALE)

# Check if image is loaded
if src_image is None:
    print("Image not found!")
    exit()

# ------------------------
# 1. Image Negation
# ------------------------
negative_image = 255 - src_image

# ------------------------
# 2. Thresholding
# ------------------------
_, thresholded_image = cv2.threshold(
    src_image, 128, 255, cv2.THRESH_BINARY
)
```

```
# ------------------------
# 3. Gamma Correction
# ------------------------
gamma = 2.0

# Normalize image to range [0, 1]
normalized_image = src_image / 255.0

# Apply gamma correction
gamma_corrected_image = np.power(normalized_image, 1 /
gamma)

# Convert back to range [0, 255]
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

# ------------------------
# Display Images
# ------------------------
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1, 4, 2)
```

```
plt.imshow(negative_image, cmap='gray')

plt.title("Negative")

plt.axis('off')


plt.subplot(1, 4, 3)

plt.imshow(thresholded_image, cmap='gray')

plt.title("Threshold")

plt.axis('off')


plt.subplot(1, 4, 4)

plt.imshow(gamma_corrected_image, cmap='gray')

plt.title("Gamma")

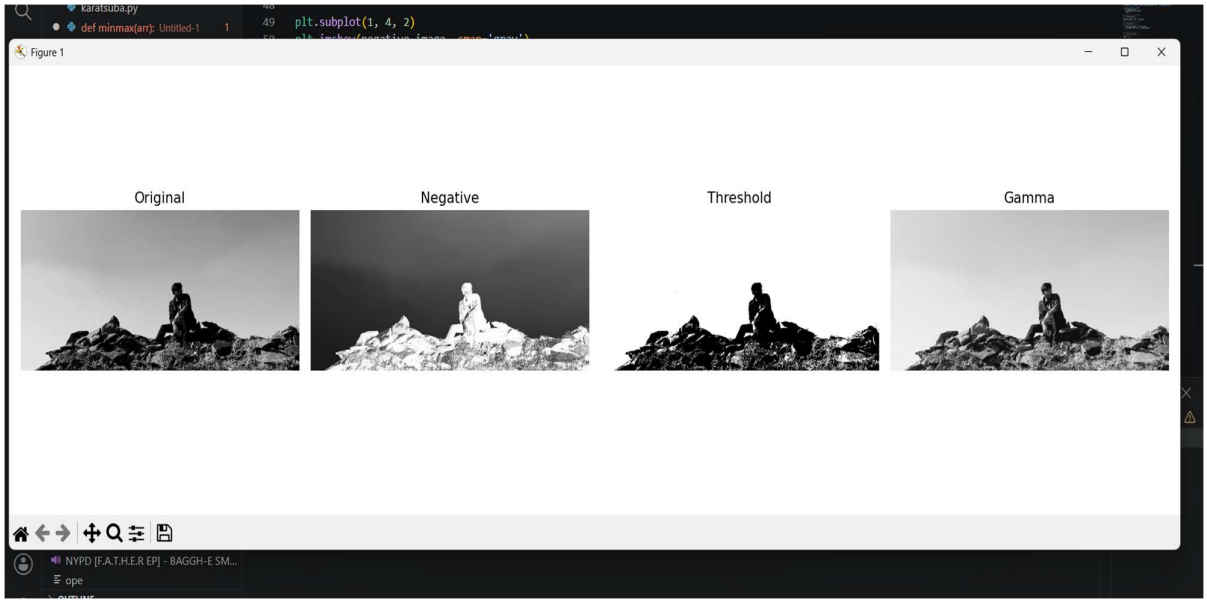
plt.axis('off')


plt.tight_layout()

plt.show()
```

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Output:



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